Lab 1A
Business Requirements Analysis
Retail Analytics Platform

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1 Introduction

This report documents the business requirements analysis for a retail sales analytics platform (*Retail Analytics*). The objective of Lab 1A is to understand the business problem, define commercial objectives, identify functional and non-functional requirements, and design a dashboard wireframe that will serve as the foundation for the ETL process to be implemented in Lab 1B [**?**].

The project team consists of three members with defined roles following agile methodologies: a Project Manager, a Product Owner, and a Development Team member [**?**].

2 Business Problem Analysis

2.1 Problem

The company has sales, inventory, and product data, but the information is scattered across spreadsheets and different systems. This makes it difficult for managers to get a quick and reliable view of store performance and make timely decisions.

2.2 Importance

Resolving this problem is important because it:

- Reduces time spent searching for information.
- Enables data-driven decision-making.
- Quickly identifies underperforming products.
- Facilitates tracking of goal compliance.
- Improves marketing campaign planning.

2.3 Stakeholders

Stakeholder	**Expected Benefit**
Regional Managers	Monitor the performance of their stores
Store Managers	Analyze sales and local trends
Marketing Team	Adjust campaigns based on each store's performance
Senior Management	Obtain strategic indicators for decision-making

Table 1: Main stakeholders and their expected benefits.

2.4 Business Decisions

With an adequate analytics platform, managers will be able to:

- Identify stores with low performance.
- Detect product categories with declining sales.
- Design promotions for specific products.
- Compare performance across regions.
- Evaluate compliance with commercial goals.
- Plan inventory based on sales trends.

2.5 Missing Information

Before designing the solution, it is necessary to ask the client:

Question	Justification
Which indicators are priorities for management?	Define the main KPIs
How often do sales goals change?	Properly interpret goal compliance
What access levels will each user have?	Define permissions and security
How many stores and regions exist?	Estimate data volume
What system stores promotions?	Properly integrate sources
What format should downloadable reports have?	Meet business needs
What exactly does a "successful store" mean?	Establish evaluation criteria
What historical data should be retained?	Define the analysis period

Table 2: Additional questions for the client.

3 Business Objectives

Business Objective	Why is it Important?
Centralize sales information into a single analytical view	Reduces time spent searching for information across different files and systems
Monitor sales performance by region and store	Allows quickly identifying branches with best and worst performance
Analyze category and product performance	Helps detect low-demand products and define promotions
Evaluate compliance with commercial goals	Helps measure objective achievement and adjust campaigns
Analyze sales trends over time	Allows comparing daily, weekly, monthly, or annual performance
Facilitate clear and easy access to information for managers	Increases platform adoption and reduces the learning curve
Analyze the impact of promotions on sales	Helps measure whether campaigns are actually driving sales
Facilitate operational tracking of transactions	Provides day-to-day indicators such as average order value

Table 3: Business objectives and their importance.

4 Analytical Questions

Business Question	Information Required
What is the total sales by day, week, month, and year?	Sales transactions with date and amount
Which product categories generate the highest and lowest revenue?	Sales grouped by product category
Which products have the best and worst performance?	Quantity sold and revenue by product
Which stores have the best and worst performance?	Total sales by store
How do sales vary across regions?	Sales grouped by region
How have sales evolved compared to the previous month or year?	Sales history by period
Which stores meet or fail to meet their commercial goals?	Actual sales and goals for each store
What impact do promotions have on sales?	Promotion information related to sales
What sales trends exist by period?	Sales time series (daily, weekly, monthly)
What reports do managers and marketing need to download?	Consolidated sales, goals, and promotions data

Table 4: Analytical questions and the information required to answer them.

5 Requirements Identification

5.1 Functional Requirements

ID	Functional Requirement	Justification
FR-01	Display total sales by day, week, month, and year	Enables commercial performance monitoring
FR-02	Visualize sales by region and by store	Facilitates comparing performance across branches
FR-03	Display category and product performance	Identifies high and low performing products
FR-04	Allow filtering information by region, store, and time period	Facilitates specific analysis according to user needs
FR-05	Display historical sales trends	Allows comparing behavior across different periods
FR-06	Display goal compliance by store	Helps commercial and marketing areas evaluate results
FR-07	Integrate promotion information with sales	Allows analyzing the impact of promotional campaigns
FR-08	Allow report downloads	Facilitates sharing information outside the platform

Table 5: Functional requirements of the system.

5.2 Non-Functional Requirements

ID	Non-Functional Requirement	Justification
NFR-01	The dashboard must update information daily	The client indicated daily updates are sufficient
NFR-02	The platform must be easy to use and intuitive	Some managers do not have technical knowledge
NFR-03	Only authorized users will be able to access the information	Protects the company's commercial data
NFR-04	Indicator loading time must not exceed 5 seconds	Improves user experience
NFR-05	The platform must be available during business hours	Ensures access when managers need it
NFR-06	It must be compatible with mobile devices	Some managers work while traveling
NFR-07	It must allow growth in number of stores and data volume	Facilitates business scalability
NFR-08	The information presented must be consistent and reliable	Prevents decisions based on incorrect data

Table 6: Non-functional requirements of the system.

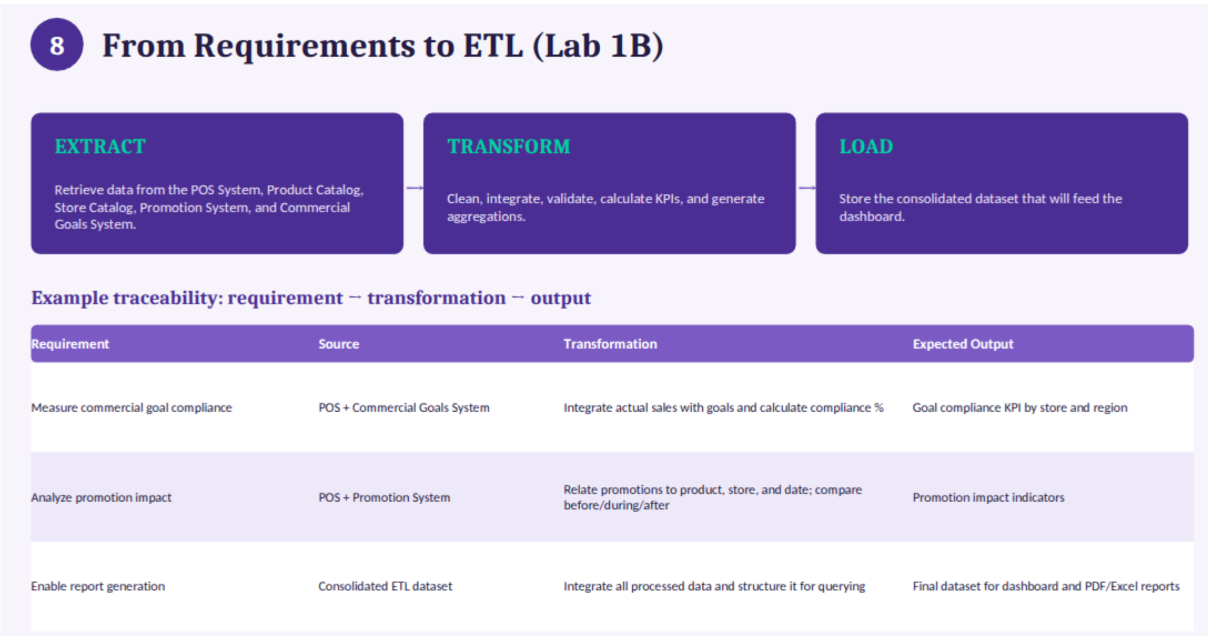

Figure 1: Summary of functional and non-functional requirements.

5.3 Data Requirements

Information Needed	Data Required	Possible Source	Why is it Needed?
Monthly Sales	Sales transactions	POS System	Calculate total sales and trends
Sales by Store	Store ID, date, and amount	POS System	Compare performance across branches
Sales by Region	Store and region information	POS + Store Catalog	Analyze geographic differences
Product Performance	Product catalog and sales	Catalog + POS	Identify successful products
Category Performance	Categories and sales	Product Catalog	Evaluate categories by revenue
Goal Compliance	Commercial goals and actual sales	Goals System	Measure objective achievement
Active Promotions	Campaigns, dates, and products	Promotion System	Analyze promotion impact on sales
Historical Trends	Sales history	POS System	Compare periods and detect patterns

Table 7: Data requirements for the analytics platform.

Information Needed	Data Required	Possible Source
Monthly sales	Sales transactions	POS System
Sales by store	Store ID, date, and amount	POS System
Sales by region	Store and region information	POS System + Store Catalog
Product performance	Product catalog and sales	Product Catalog + POS System
Category performance	Categories and sales	Product Catalog
Goal compliance	Commercial goals and actual sales	Commercial Goals System
Active promotions	Campaigns, dates, and products	Promotion System
Historical trends	Sales history	POS System

Figure 2: Data requirements table with sources and transformations.

6 User Stories

6.1 Story 1

As a regional manager, **I want to** visualize sales of all stores in my region, **so that** I can quickly identify which ones have the best and worst performance.

6.2 Story 2

As a store manager, **I want to** view sales by product category, **so that** I can detect which products need promotions or commercial strategies.

6.3 Story 3

As a marketing team member, **I want to** visualize which stores meet their commercial goals, **so that** I can adjust advertising campaigns and strengthen underperforming stores.

6.4 Story 4

As a regional manager, **I want to** compare current sales with those from the previous month and year, **so that** I can evaluate growth or decline in commercial performance.

6.5 Story 5

As a commercial director, **I want to** download consolidated reports of sales and goal compliance, **so that** I can present them in follow-up meetings and support strategic decision-making.

6.6 Story 6 (Added Value)

As a store manager, **I want to** visualize the impact of promotions on sales, **so that** I can identify which campaigns generate the best results.

User	Decision to Make	Information Needed
Regional Manager	Identify stores with best and worst performance	Sales by region and by store
Store Manager	Decide which products need promotion	Sales by product and category
Marketing Team	Adjust commercial campaigns	Goal compliance and sales results
Commercial Director	Evaluate overall company performance	KPIs, trends, and historical comparisons
Regional Manager	Detect changes in sales behavior	Daily, weekly, and monthly trends
Store Manager	Evaluate promotion effectiveness	Promotion data and associated sales

Table 8: Decision and information needs by user role.

7 KPIs

Business Objective	KPI	Justification
Centralize sales information	Total Revenue	Measures the total value of sales made
Monitor store performance	Sales by Store	Allows comparing performance across branches
Analyze performance by region	Sales by Region	Facilitates geographic analysis
Evaluate categories and products	Sales by Category	Identifies categories with highest contribution
Evaluate categories and products	Top 10 Best-Selling Products	Knows the most in-demand products
Analyze sales trends	Sales Growth (%)	Compares sales with previous periods
Evaluate goal compliance	Goal Compliance Percentage	Measures the degree of objective achievement
Analyze promotion impact	Sales Increase by Promotion	Evaluates impact of promotional campaigns
Operational tracking	Average Order Value	Average value of each purchase made

Table 9: KPIs aligned with business objectives.

7.1 KPIs requiring data from multiple systems

The following KPIs require integrating information from more than one source:

KPI	Systems Involved
Goal Compliance Percentage	POS System + Commercial Goals System
Sales Increase by Promotion	POS System + Promotion System

Table 10: KPIs requiring integration of multiple data sources.

7.2 Most difficult KPI to calculate

The **Sales Increase by Promotion** would be the most complex KPI because:

- It requires integrating sales and promotion data from different systems.
- Each promotion must be related to the corresponding products, stores, and dates.
- To measure the impact, sales before, during, and after each campaign must be compared, avoiding attributing changes to external factors such as seasonality.

8 Dashboard Wireframe

The lab asks to design a low-fidelity wireframe of the dashboard. The proposal includes:

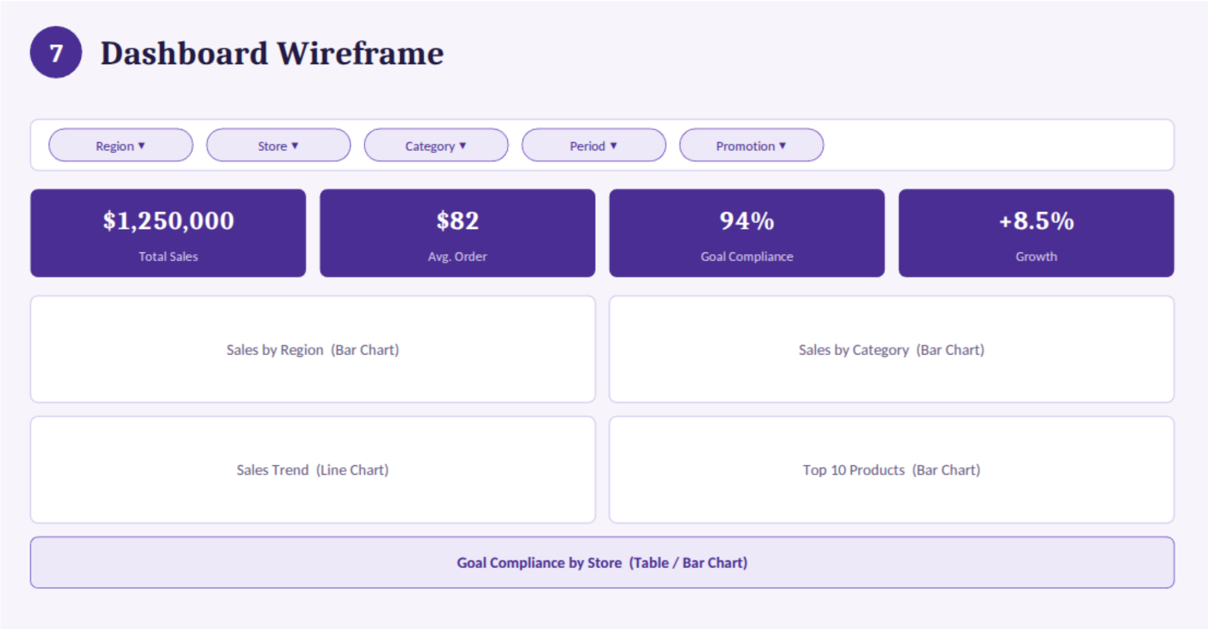

Figure 3: Proposed sales analytics dashboard wireframe.

8.1 Design Justification

Filters: Located at the top because they affect all displayed information. They include region, store, category, period, and promotion, responding to the client's requirements to analyze information from different perspectives.

KPIs: The first row presents the most important indicators: total sales, average order value, goal compliance, and growth percentage. These are metrics that managers need to check immediately.

Charts: Four main visualizations:

- Bars by Region: Compare performance across regions.
- Bars by Category: Identify categories with best and worst performance.
- Trend Line: Analyze sales evolution over time.
- Top 10 Products: Identify products with the highest sales volume.

Goal Compliance: A separate chart or table because the client indicated that the marketing area needs to evaluate which stores meet their commercial objectives.

Export: A section for exporting reports in PDF or Excel, since the client mentioned the need to download reports.

The dashboard follows a logical reading flow: filters (define the context) → KPIs (overview) → charts (deep analysis) → details (goal compliance) → export (share information) [?].

9 Preparing for Lab 1B

Business Requirement	Data Source	Transformation Needed	Expected Output
Visualize total sales by period	POS System	Clean records, convert dates, calculate totals	Dataset with sales aggregated by period
Analyze sales by region	POS + Store Catalog	Join tables by store ID, group by region	Sales consolidated by region
Compare store performance	POS + Store Catalog	Relate sales with store information	Metrics per branch
Analyze product/category performance	POS + Product Catalog	Relate products with categories	Indicators by category and product
Measure goal compliance	POS + Goals System	Integrate actual sales with assigned goals	Goal compliance KPI by store
Analyze promotion impact	POS + Promotion System	Relate promotions with sales before/during/after	Impact indicators
Analyze historical trends	POS (historical)	Sort chronologically, generate aggregations	Time series for charts
Enable report generation	Consolidated ETL dataset	Integrate all processed data	Final dataset for dashboard and reports

Table 11: Business requirements mapped to the ETL process for Lab 1B.

9.1 Transformation Justification

The proposed transformations correspond to the classic stages of an ETL process [?]:

Transformation	Objective
Data Cleaning	Remove incomplete, duplicate, or inconsistent records
Format Conversion	Standardize dates, currencies, and data types
Source Integration	Combine information from POS, catalogs, goals, and promotions
Aggregation	Calculate summary metrics by period, region, or store
KPI Calculation	Generate metrics ready for dashboard consumption

Table 12: ETL transformations and their objectives.

9.2 Relationship with the ETL Process

ETL	Application in This Case
Extract	Retrieve data from POS System, Product Catalog, Store Catalog, Promotion System, and Commercial Goals System
Transform	Clean, integrate, validate, calculate KPIs, and generate aggregations
Load	Store the consolidated dataset that will feed the analysis dashboard

Table 13: Application of ETL stages to the case study.

10 Reflection

10.1 Why is a Dashboard not the first step?

No. This activity showed that building a dashboard is actually one of the last visible steps, not the first one. Before any table, chart, or wireframe can be designed, the team had to understand the business problem, identify stakeholders, translate their needs into objectives and analytical questions, and only then define functional, non-functional, and data requirements.

This prior work made clear that the data sources involved (POS System, Product Catalog, Store Catalog, Commercial Goals System, and Promotion System) need to be integrated before any KPI can be trusted. If the team had started by designing the dashboard, it likely would have missed requirements such as mobile access, security levels, or the need to relate promotions to sales.

In short, the dashboard is the visible outcome of the process, but the real first step is understanding the business problem and the data needed to solve it [**?**].

11 AI Prompts

As part of this activity, the team used an AI assistant as a support tool to organize ideas, check for completeness, and refine the wording of the deliverables. All content was reviewed, validated, and adjusted by the team before being included in this report.

Activity	Prompt Purpose	Example Prompt
General	Translate and clarify instructions	"Translate this lab statement into clear, simple terms and list what each activity is asking for"
Activity 1	Identify gaps in requirements	"Based on this client transcript, what information is missing that a data engineer should ask about?"
Activity 2	Turn needs into objectives	"Help me phrase these client needs as business objectives, separate from any technical solution"
Activity 3	Convert objectives into questions	"For each business objective, suggest a business question and the information required to answer it"
Activity 4	Distinguish requirement types	"Given these business objectives, suggest functional, non-functional, and data requirements, and explain the difference"
Activity 5	Draft user stories	"Write user stories in the As a/I want/So that format for a regional manager, store manager, and marketing member"
Activity 6	Define and justify KPIs	"Suggest KPIs aligned with these business objectives, and identify which ones require combining data from more than one system"
Activity 7	Structure the wireframe	"Suggest a logical layout for a sales dashboard wireframe, including filters, KPIs, charts, and an export section"
Review	Check consistency	"Review whether these business objectives, KPIs, and requirements are consistent with each other"

Table 14: AI prompts used during the lab development.

12 Conclusions

Lab 1A allowed a complete analysis of business requirements for the retail sales analytics platform. The business problem, stakeholders, commercial objectives, analytical questions, functional and non-functional requirements, user stories, KPIs, and a dashboard wireframe were clearly identified.

It is concluded that the integration of multiple data sources (POS System, Product Catalog, Store Catalog, Commercial Goals System, and Promotion System) is a fundamental requirement for the platform to answer all the business questions raised. The ETL process to be implemented in Lab 1B will be the mechanism to achieve this integration.